

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

AI MINI PROJECT

MAYUR STATIONERY – BUSINESS ANALYTICS PROJECT

Student Name	Asmitha Mohan Raj
Roll No.	T002
Class / Batch	TYIT / 1
Subject	AI and Application Development and Jira
Academic Year	2026 - 2027

Submitted to: Prof. Chetana Tanavade

1. Introduction:

Business Analytics is the process of using data, statistical methods, visualization techniques and analytical models to support better business decisions. It helps organizations understand past performance, identify patterns, predict future trends and improve business operations.

In this project, Business Analytics techniques are applied to the sales data of Mayur Stationery. The dataset contains information related to products, units sold, prices, discounts, customers, categories, store locations and payment methods.

The main purpose of this project is to analyze the sales performance of Mayur Stationery and identify meaningful business patterns. Various analytical techniques such as Exploratory Data Analysis, descriptive statistics, correlation, regression, probability, sampling, confidence intervals, hypothesis testing, time series forecasting, optimization and ANOVA are used.

The results obtained from the analysis can help the business improve inventory planning, understand customer and product behavior, evaluate promotional strategies and make data-driven business decisions.

2. Problem Statement:

Mayur Stationery generates sales transactions across different products, categories, store locations and payment methods. However, raw sales data alone does not provide sufficient information about product performance, customer patterns, sales trends and future demand.

The problem is to analyze the available stationery sales data using Business Analytics techniques and identify useful patterns and relationships.

The project aims to determine which products and categories perform better, how sales vary over time, whether important variables are related, what future sales may look like and how limited business resources can be utilized efficiently.

The analysis will provide useful information for inventory planning, pricing, promotional activities, sales management and business decision-making.

3. Objectives:

The main objectives of the project are:

1. To collect and prepare the Mayur Stationery sales dataset for analysis.
2. To perform Exploratory Data Analysis to understand the structure and characteristics of the dataset.
3. To analyze sales patterns, customer characteristics, product categories and store performance.
4. To identify relationships between important numerical variables using correlation and regression analysis.
5. To apply probability, sampling, confidence intervals and hypothesis testing techniques.

6. To analyze historical sales trends and forecast future sales using time series forecasting.
7. To determine an optimal production strategy using an optimization model under limited machine and labour resources.
8. To derive meaningful business insights and recommendations for better decision-making.

4. Business Scenario / Domain Description:

4.1 Business Name : Mayur Stationery

4.2 Business Type

Mayur Stationery is a stationery retail business that sells different stationery and office-related products to customers.

4.3 Products and Services

The business deals with stationery products such as writing materials, school supplies, office supplies and other stationery-related products.

4.4 Target Customers

The target customers may include students, teachers, office employees, professionals and general customers who purchase stationery products.

4.5 Business Operations

The business records transaction-level information including products purchased, units sold, unit prices, discounts, customer details, categories, store locations and payment methods.

4.6 Business Problems

The business needs to understand product demand, category performance, store performance, customer characteristics, discount usage and sales trends. It also needs to estimate future sales and use available resources efficiently.

Business Analytics can help convert the available sales data into useful information for planning and decision-making.

5. Dataset Description:

The dataset used for this project is **Mayur Stationery Sales Analysis.csv**.

The dataset contains **1,000 sales transaction records** and **15 original columns**. It includes information about transactions, products, customers, prices, discounts, categories, locations and payment methods.

5.1 Dataset Information:

Parameter	Details
Dataset Name	Mayur Stationery Sales Analysis
File Type	CSV
Number of Records	1,000
Number of Columns	15
Analysis Tool	Python / Google Colab
Main Library	Pandas
Visualization	Matplotlib, Seaborn, Plotly
Statistical Analysis	NumPy, SciPy
Machine Learning	Scikit-learn

5.2 Dataset Variables:

Column	Description	Data Type
Date	Date of transaction	Date
Product	Name of product purchased	Categorical
Units_Sold	Number of units sold	Numerical
Unit_Price	Price per unit	Numerical
Transaction_ID	Unique transaction identifier	ID

Customer_ID	Unique customer identifier	ID
Customer_Name	Name of customer	Categorical
Category	Product category	Categorical
Store_Location	Location of store	Categorical
Payment_Method	Payment method used	Categorical
Discount_Applied	Discount percentage applied	Numerical
Net_Price	Price after discount	Numerical
Day_of_Week	Day of transaction	Categorical
Month	Month of transaction	Categorical
Customer_Age	Age of customer	Numerical

Figure 1: Sample of Mayur Stationery Sales Dataset

Dataset uploaded successfully! Number of records: 999 Number of columns: 15										
	Date	Product	Units_Sold	Unit_Price	Transaction_ID	Customer_ID	Customer_Name	Category	Store_Location	Payment_Method
0	13-04-2025	Paper	11	36.34	T000001	C2202	Customer 1825	Accessories	Downtown	UPI
1	29-06-2025	Notebook	90	18.66	T000002	C0596	Customer 410	Paper	Downtown	Cash
2	03-04-2025	Notebook	15	45.98	T000003	C1744	Customer 4507	Accessories	Downtown	Card
3	15-01-2025	Stapler	24	29.72	T000004	C1863	Customer 4013	Accessories	Mall	Card
4	17-04-2025	Stapler	92	4.02	T000005	C8452	Customer 3658	Paper	Downtown	Cash

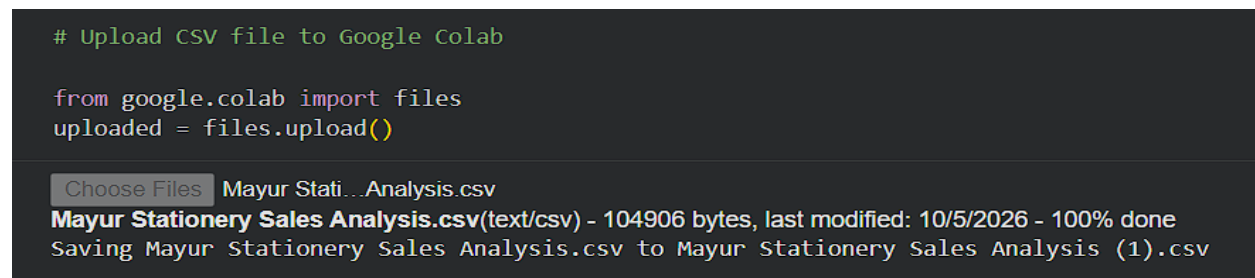
6. Data Collection

The Mayur Stationery sales dataset was used for the Business Analytics project. The dataset consists of 1,000 transaction records containing information about products, customers, prices, discounts, categories, store locations and payment methods.

The CSV file was imported into Google Colab using Python and the Pandas library. After loading the dataset, its structure and contents were examined before performing further analysis.

The dataset contains transaction information for the available period from January 2025 to August 2025.

Figure 2: Dataset Loading and Initial Inspection

A screenshot of the Google Colab interface. At the top, there is a code editor with the following Python code:

```
# Upload CSV file to Google Colab

from google.colab import files
uploaded = files.upload()
```

Below the code editor, there is a file upload section. It shows a button labeled "Choose Files" followed by the text "Mayur Stati...Analysis.csv". Below this, it says "Mayur Stationery Sales Analysis.csv(text/csv) - 104906 bytes, last modified: 10/5/2026 - 100% done" and "Saving Mayur Stationery Sales Analysis.csv to Mayur Stationery Sales Analysis (1).csv".

```
# Upload CSV file to Google Colab

from google.colab import files
uploaded = files.upload()

Choose Files Mayur Stati...Analysis.csv
Mayur Stationery Sales Analysis.csv(text/csv) - 104906 bytes, last modified: 10/5/2026 - 100% done
Saving Mayur Stationery Sales Analysis.csv to Mayur Stationery Sales Analysis (1).csv
```

7. Data Preprocessing:

Data preprocessing was performed to ensure that the dataset was suitable for analysis. The dataset was checked for missing values, duplicate records, incorrect data types and inconsistent values.

The **Date** column was converted into the appropriate date format. Additional business variables were also created to support sales analysis.

7.1 Preprocessing Steps

1. Loaded the CSV dataset.
2. Checked the number of rows and columns.
3. Checked column names and data types.
4. Checked for missing values.
5. Checked for duplicate records.
6. Converted the Date column into datetime format.
7. Removed duplicate records if present.
8. Calculated Gross Sales.
9. Calculated Discount Amount.
10. Calculated Final Sales.

7.2 Calculated Variables

Gross Sales : $\text{Gross Sales} = \text{Units Sold} \times \text{Unit Price}$

Discount Amount : $\text{Discount Amount} = \text{Gross Sales} \times \text{Discount Percentage} / 100$

Final Sales : $\text{Final Sales} = \text{Gross Sales} - \text{Discount Amount}$

7.3 Data Quality Results

- Total records: **1,000**
- Missing values: **0**
- Duplicate records: **0**
- Records after cleaning: **1,000**

Figure 3: Data Information using df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 999 entries, 0 to 998
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  999 non-null   object
1   Product               999 non-null   object
2   Units_Sold            999 non-null   int64
3   Unit_Price            999 non-null   float64
4   Transaction_ID        999 non-null   object
5   Customer_ID           999 non-null   object
6   Customer_Name         999 non-null   object
7   Category              999 non-null   object
8   Store_Location        999 non-null   object
9   Payment_Method        999 non-null   object
10  Discount_Applied      999 non-null   float64
11  Net_Price             999 non-null   float64
12  Day_of_Week           999 non-null   object
13  Month                 999 non-null   object
14  Customer_Age          999 non-null   int64
dtypes: float64(3), int64(2), object(10)
memory usage: 117.2+ KB
```

Figure 4: Missing Value Analysis

```
Missing Values:
Date                0
Product             0
Units_Sold          0
Unit_Price          0
Transaction_ID      0
Customer_ID         0
Customer_Name       0
Category            0
Store_Location      0
Payment_Method      0
Discount_Applied    0
Net_Price           0
Day_of_Week         0
Month               0
Customer_Age        0
dtype: int64
```

8. Exploratory Data Analysis:

Exploratory Data Analysis (EDA) was performed to understand the characteristics, distribution and patterns present in the dataset.

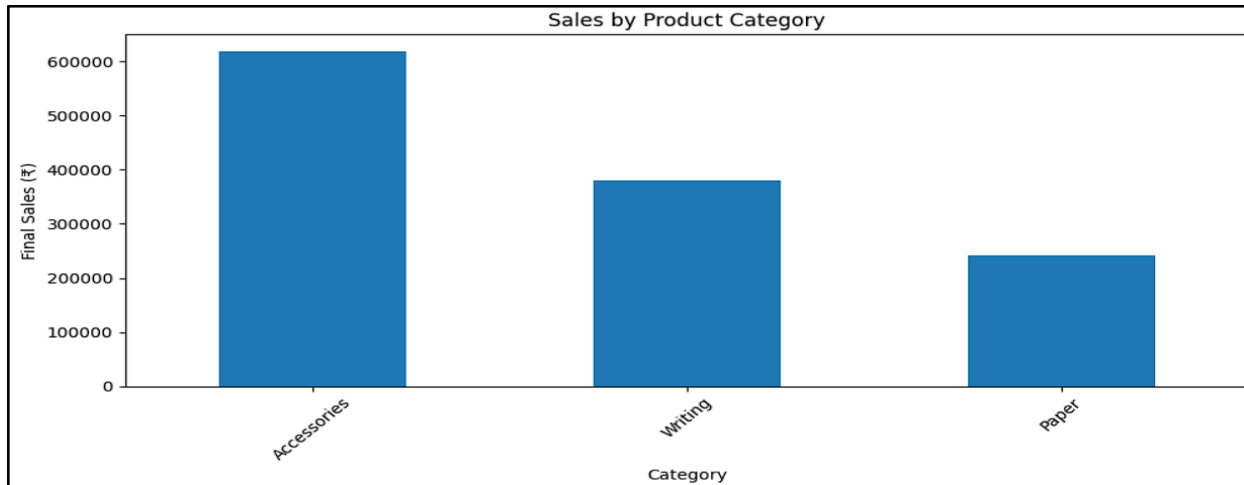
The analysis included examination of the first and last records, dataset dimensions, data types, missing values, duplicate values, unique values and descriptive statistics.

Different visualizations were also created to understand sales performance and customer behavior.

8.1 Bar Chart – Sales by Category:

The sales performance of different product categories was visualized using a bar chart.

Figure 5: Sales by Category

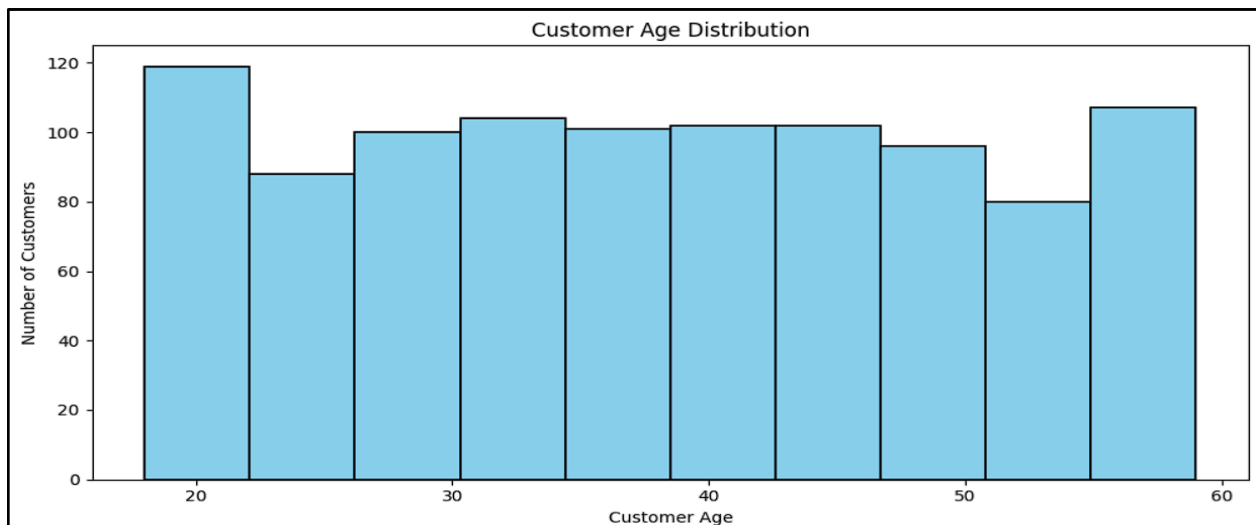


Observation: The chart compares the sales contribution of different product categories and helps identify categories with relatively higher and lower sales.

8.2 Histogram – Customer Age:

A histogram was used to understand the distribution of customer ages.

Figure 7: Customer Age Distribution

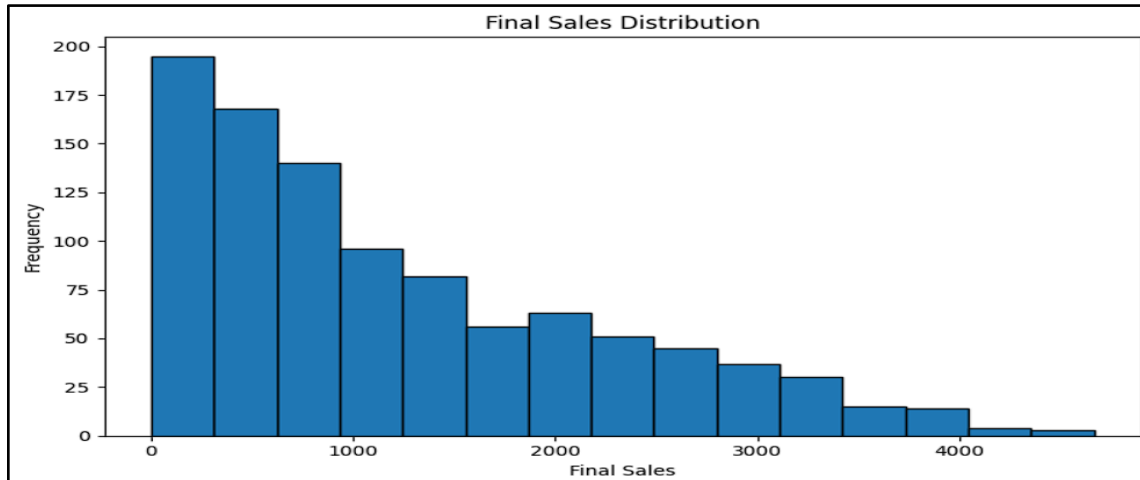


Observation: The histogram shows the age distribution of customers and indicates the age ranges in which most customers are concentrated.

8.3 Histogram – Final Sales:

A histogram was created for Final Sales.

Figure 8: Final Sales Distribution

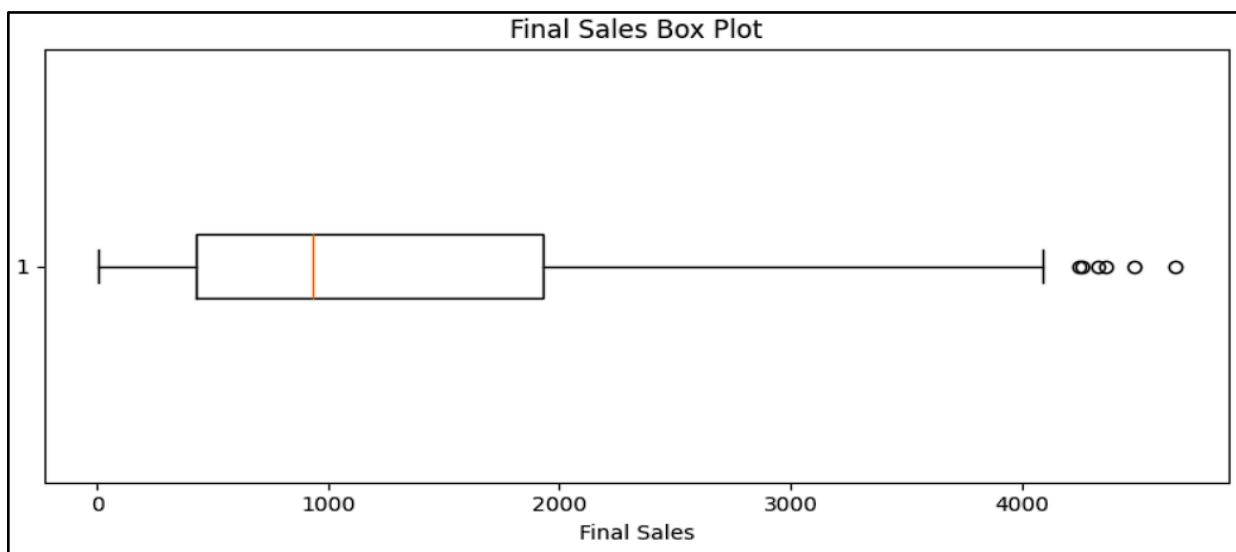


Observation: The histogram shows the distribution and spread of transaction sales values.

8.4 Box Plot – Final Sales

A box plot was used to examine the spread and possible outliers in Final Sales.

Figure 9: Final Sales Box Plot

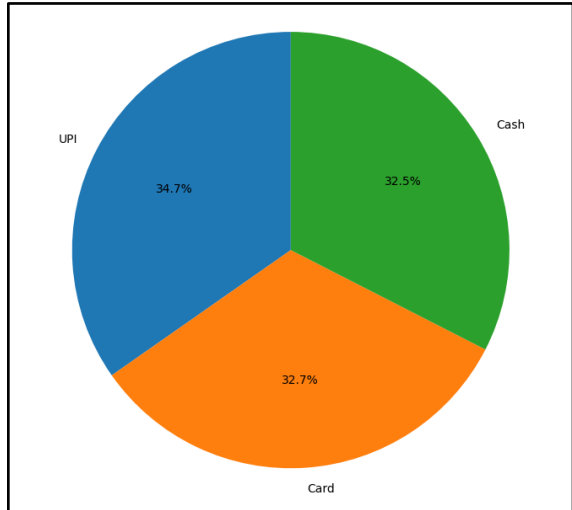


Observation: The box plot represents the median, spread and possible extreme values in transaction sales.

8.5 Pie Chart – Payment Method

A pie chart was created to show the proportion of transactions using different payment methods.

Figure 10: Payment Method Distribution

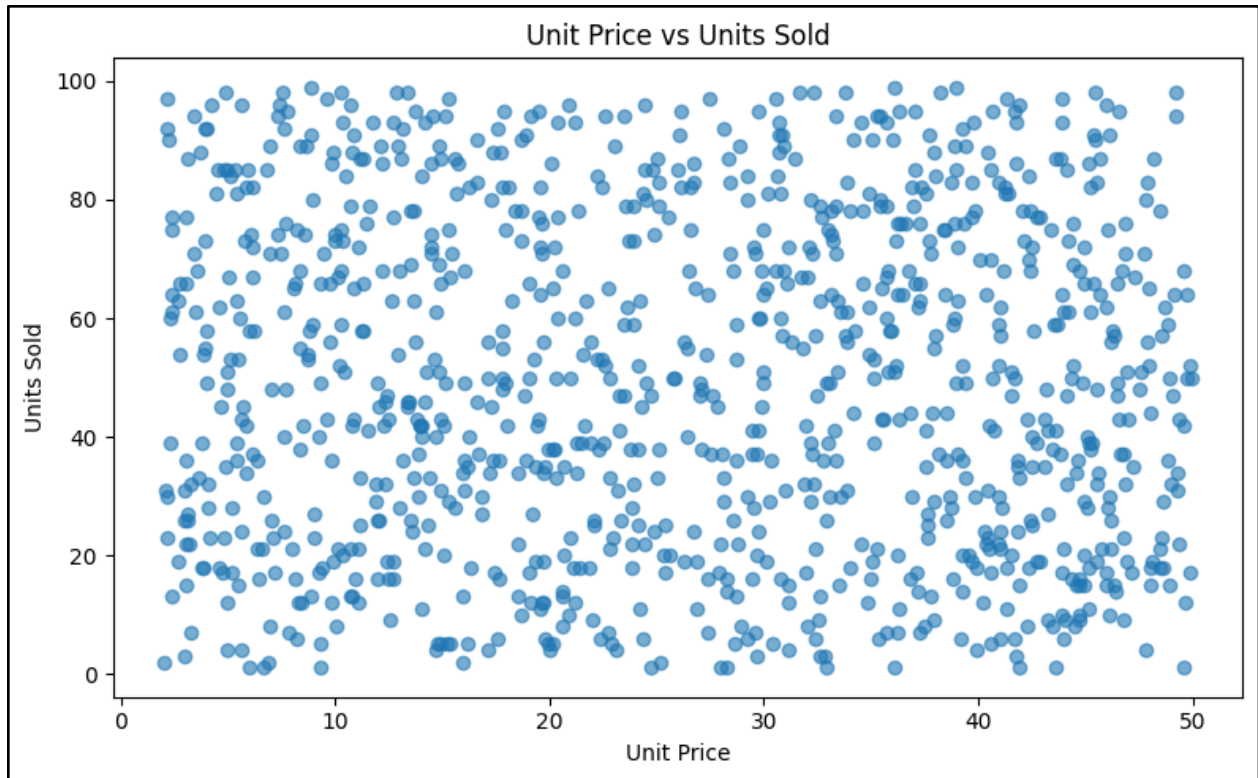


Observation: The chart shows the relative usage of different payment methods by customers.

8.6 Scatter Plot – Unit Price vs Units Sold

A scatter plot was used to examine the relationship between Unit Price and Units Sold.

Figure 11: Unit Price vs Units Sold



Observation: The scatter plot helps identify whether changes in unit price are associated with changes in the number of units sold.

9. Descriptive Statistical Analysis:

Descriptive statistics were calculated to summarize the numerical variables in the dataset. Measures of central tendency and dispersion were used to understand the characteristics of the data.

The main measures considered were mean, median, mode, minimum, maximum, range, variance and standard deviation.

9.1 Variables Analyzed

- Units Sold
- Unit Price
- Discount Applied
- Customer Age
- Final Sales

9.2 Descriptive Statistics Table

Variable	Mean	Median	Mode	Minimum	Maximum	Range	Variance	Standard Deviation
Units Sold	49.31	49.00	—	1.00	99.00	98.00	780.81	27.94
Unit Price	26.16	26.54	—	2.02	49.92	47.90	198.40	14.09
Discount Applied	2.53	2.58	—	0.00	4.99	4.99	1.98	1.41
Customer Age	38.11	38.00	—	18.00	59.00	41.00	139.31	11.80
Final Sales	—	—	—	—	—	—	—	—

The descriptive statistics provide a summary of the central value, spread and variation of the major numerical variables.

Figure 12: Descriptive Statistics Output

index	Units_Sold	Unit_Price	Discount_Applied
count	999.0	999.0	999.0
mean	49.306306306306304	26.15968968968969	2.5312812812812813
std	27.942975683530413	14.085455118511256	1.40548858276135
min	1.0	2.02	0.0
25%	24.0	13.71	1.32
50%	49.0	26.54	2.58
75%	73.5	38.84	3.72
max	99.0	49.92	4.99

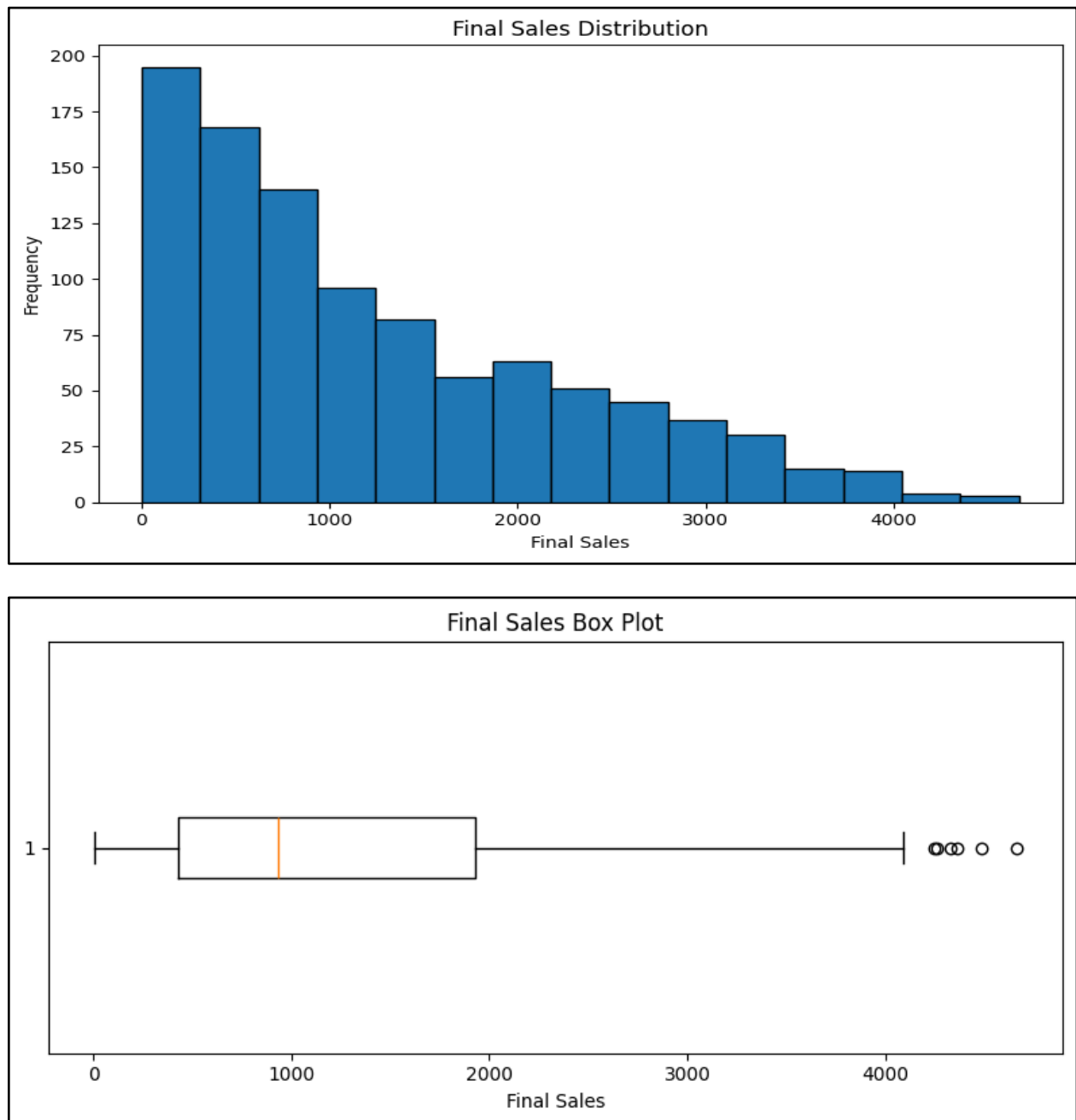
10. Distribution Analysis:

Distribution analysis was performed to understand the shape, central tendency, spread and possible outliers of the Final Sales variable.

A histogram was used to observe the frequency distribution, while a box plot was used to examine the spread and identify possible extreme observations.

The distribution analysis provides a better understanding of the typical transaction value and variation in sales.

Figure 13: Distribution of Final Sales



Observation: The distribution shows how transaction sales values are spread across the dataset and helps identify concentration and extreme observations.

11. Correlation Analysis:

Correlation analysis was performed to identify the strength and direction of relationships between numerical variables.

The variables considered include:

- Units Sold
- Unit Price
- Discount Applied
- Customer Age
- Final Sales

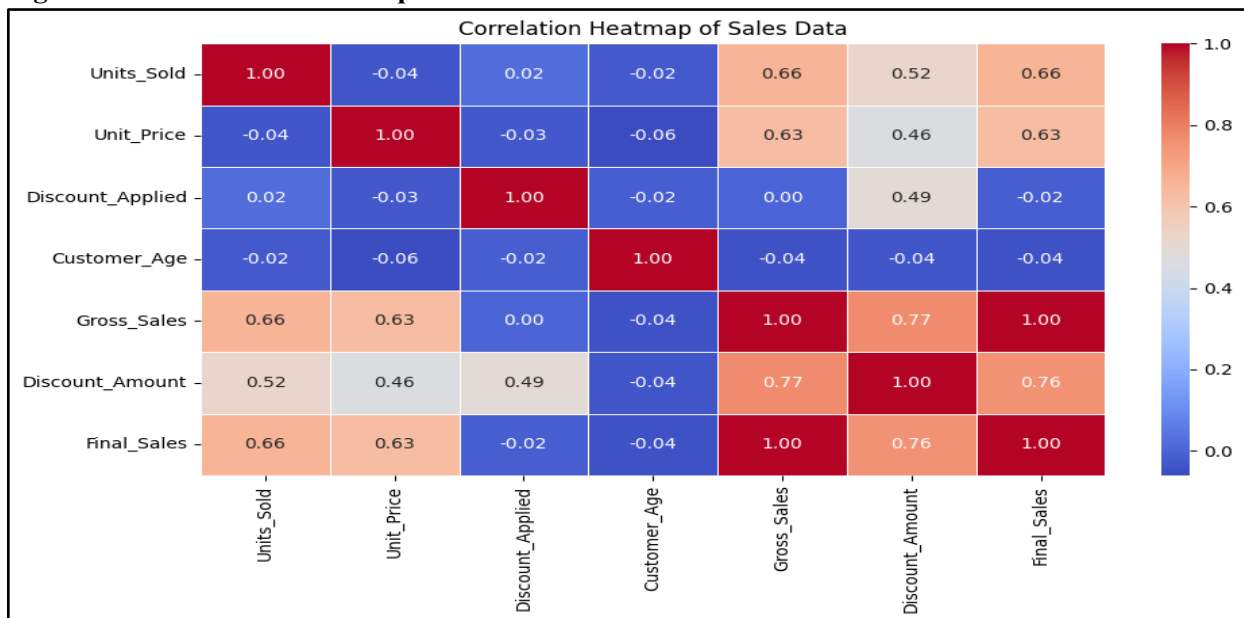
A correlation matrix and heatmap were used to visualize the relationships.

Figure 14: Correlation Matrix

Correlation Matrix:

	Units_Sold	Unit_Price	Discount_Applied	Customer_Age	Gross_Sales	Discount_Amount	Final_Sales
Units_Sold	1.000000	-0.037461	0.021370	-0.023494	0.659956	0.523211	0.659264
Unit_Price	-0.037461	1.000000	-0.030210	-0.059995	0.628112	0.459444	0.628772
Discount_Applied	0.021370	-0.030210	1.000000	-0.016853	0.001294	0.494539	-0.015600
Customer_Age	-0.023494	-0.059995	-0.016853	1.000000	-0.044305	-0.036952	-0.044196
Gross_Sales	0.659956	0.628112	0.001294	-0.044305	1.000000	0.769174	0.999761
Discount_Amount	0.523211	0.459444	0.494539	-0.036952	0.769174	1.000000	0.755011
Final_Sales	0.659264	0.628772	-0.015600	-0.044196	0.999761	0.755011	1.000000

Figure 15: Correlation Heatmap



Observation: The correlation analysis helps identify variables that have stronger positive or negative relationships with each other and with Final Sales.

12. Regression Analysis:

Regression analysis was performed to study the relationship between Units Sold and selected explanatory variables.

12.1 Dependent Variable : Units Sold

12.2 Independent Variables

- Unit Price
- Discount Applied
- Customer Age

The regression model was used to understand how these variables are associated with Units Sold and to generate predicted values.

12.3 Regression Model

Units Sold = Intercept + β_1 (Unit Price) + β_2 (Discount Applied) + β_3 (Customer Age)

12.4 Evaluation Measures

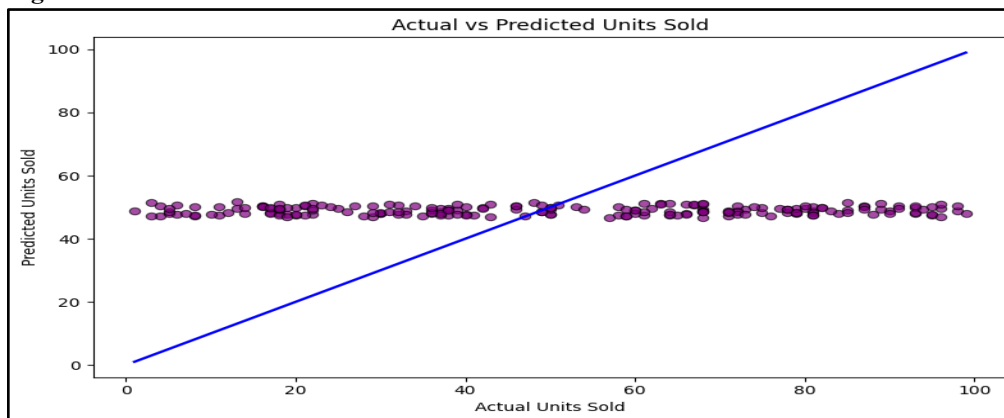
The regression model was evaluated using:

- R^2
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

Figure 16: Regression Output

```
----- Regression Results -----  
R2 Score: -0.001  
Mean Absolute Error (MAE): 24.49  
Root Mean Squared Error (RMSE): 27.87
```

Figure 17: Actual vs Predicted Units Sold



Observation: The regression results show how the selected independent variables contribute to predicting Units Sold.

13. Business Intelligence Dashboard:

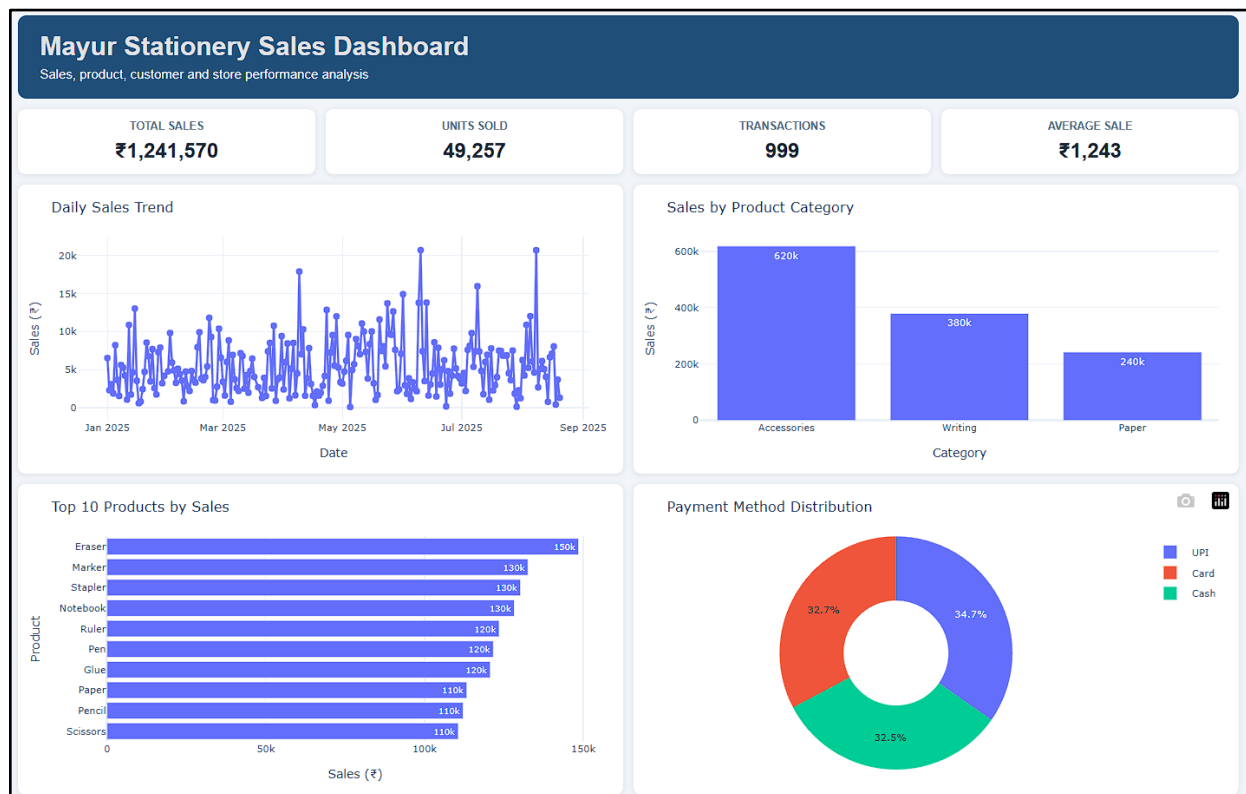
A Business Analytics dashboard was developed using Python and Plotly to present the major findings in an interactive and visually understandable form.

13.1 Dashboard Components

The dashboard includes:

- Total Final Sales
- Total Units Sold
- Average Transaction Sales
- Total Transactions
- Monthly Sales Trend
- Sales by Category
- Sales by Store Location
- Sales by Payment Method
- Top 10 Products
- Customer Age Distribution

Figure 18: Mayur Stationery Business Analytics Dashboard



13.2 Dashboard Insights

1. The dashboard provides an overall view of sales performance.
2. Monthly sales visualization helps identify changes in sales over time.
3. Category analysis helps compare product groups.

4. Store-level analysis helps identify differences between locations.
5. Top-product analysis helps identify products that contribute strongly to sales.

14. Probability Analysis:

Probability analysis was performed to understand the likelihood of different events occurring within the sales dataset.

Probabilities were calculated for product categories, payment methods and discounted transactions.

14.1 Category Probability

The probability of selecting a particular product category was calculated as:

Probability = Number of transactions in category / Total transactions

14.2 Payment Method Probability

The probability of each payment method was calculated based on its frequency in the dataset.

14.3 Discount Probability

The probability of a transaction receiving a discount was calculated.

The result obtained was:

Probability of a discounted transaction = 99.8%

This indicates that discounts were applied to almost all transactions in the dataset.

Figure 19: Category Probability

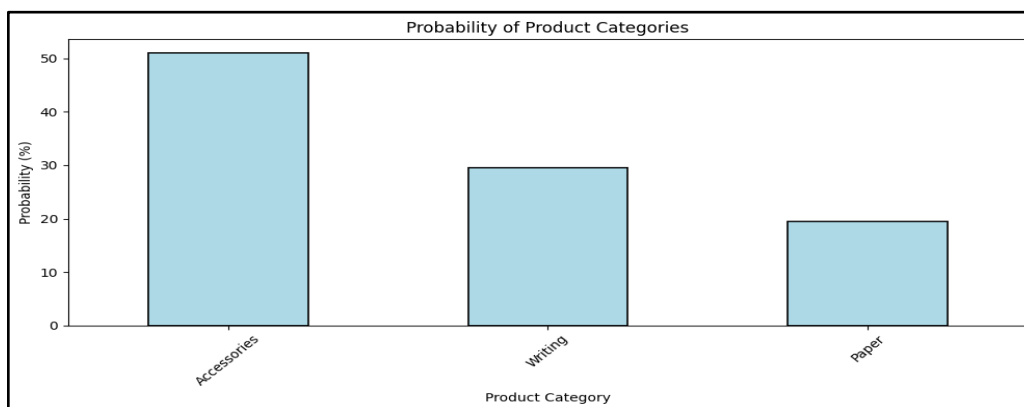
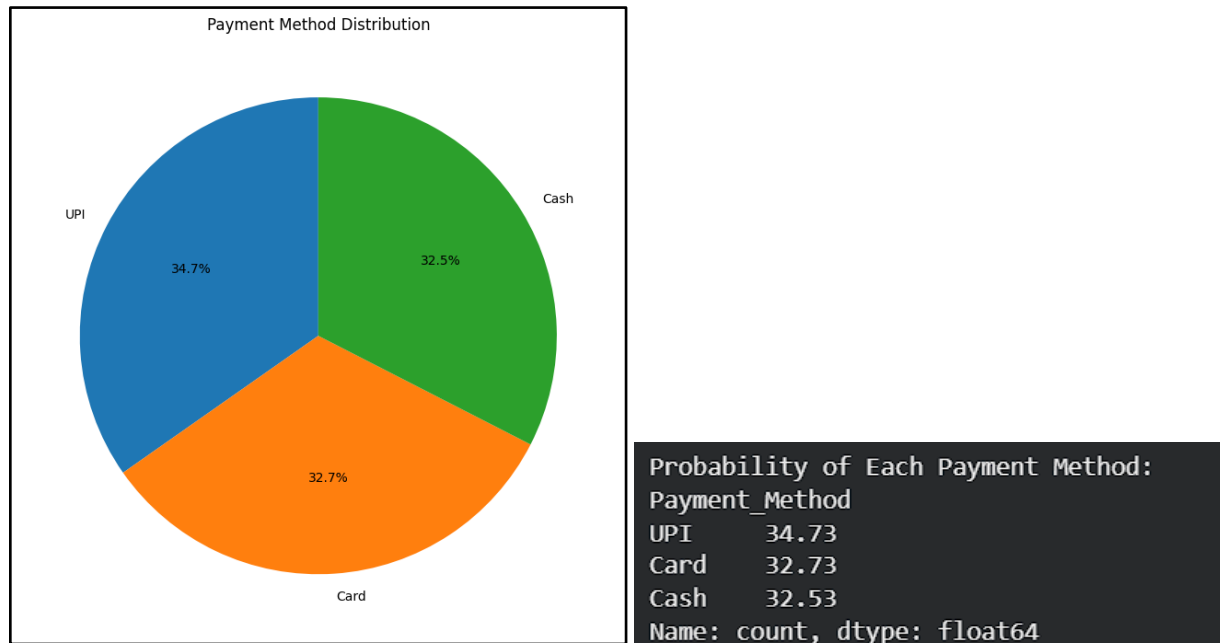


Figure 20: Payment Method Probability



15. Decision Making Under Uncertainty:

Decision-making under uncertainty involves selecting the most suitable business alternative when future outcomes are uncertain.

For Mayur Stationery, a simple decision scenario can be considered for planning inventory based on expected demand.

15.1 Alternatives

- Maintain Low Inventory
- Maintain Medium Inventory
- Maintain High Inventory

15.2 Possible Outcomes

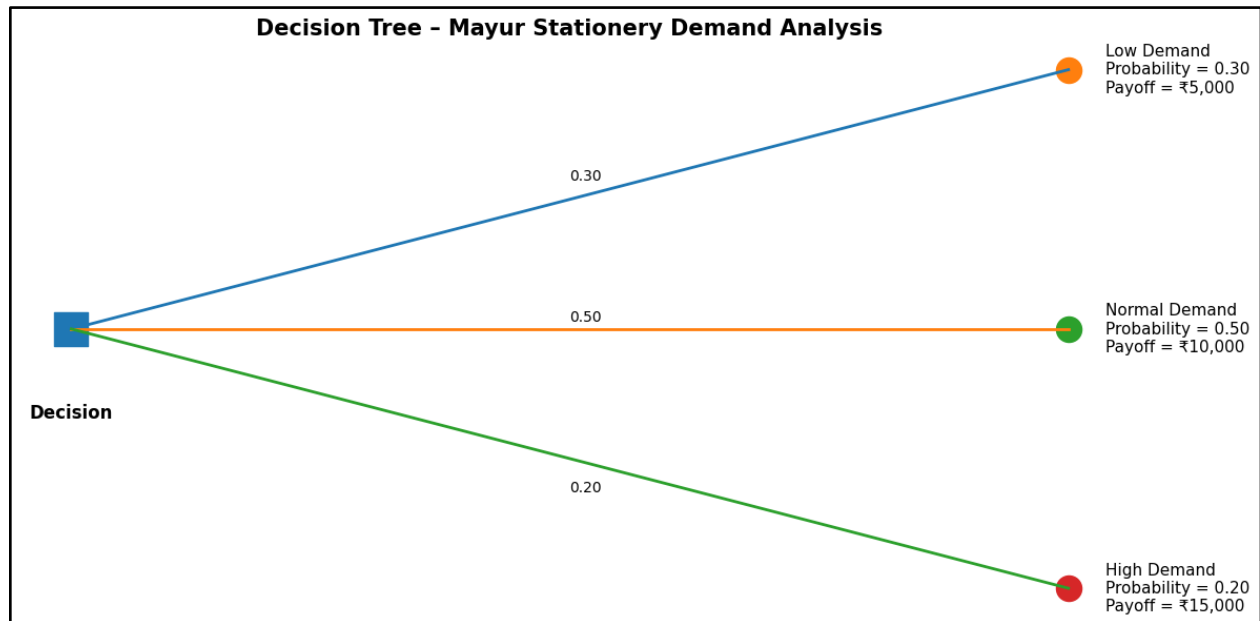
- Low Demand
- Normal Demand
- High Demand

The possible demand conditions are assigned probabilities based on historical or assumed business conditions. Expected monetary value can then be calculated for each alternative.

Expected Value = $\sum (\text{Probability} \times \text{Payoff})$

The alternative with the highest expected value can be selected as the preferred decision.

Figure 21: Decision Tree



Observation: The decision tree provides a structured method for comparing alternatives under uncertain demand conditions.

16. Sampling and Sampling Distribution:

Sampling was performed by selecting observations from the population of 1,000 sales transactions.

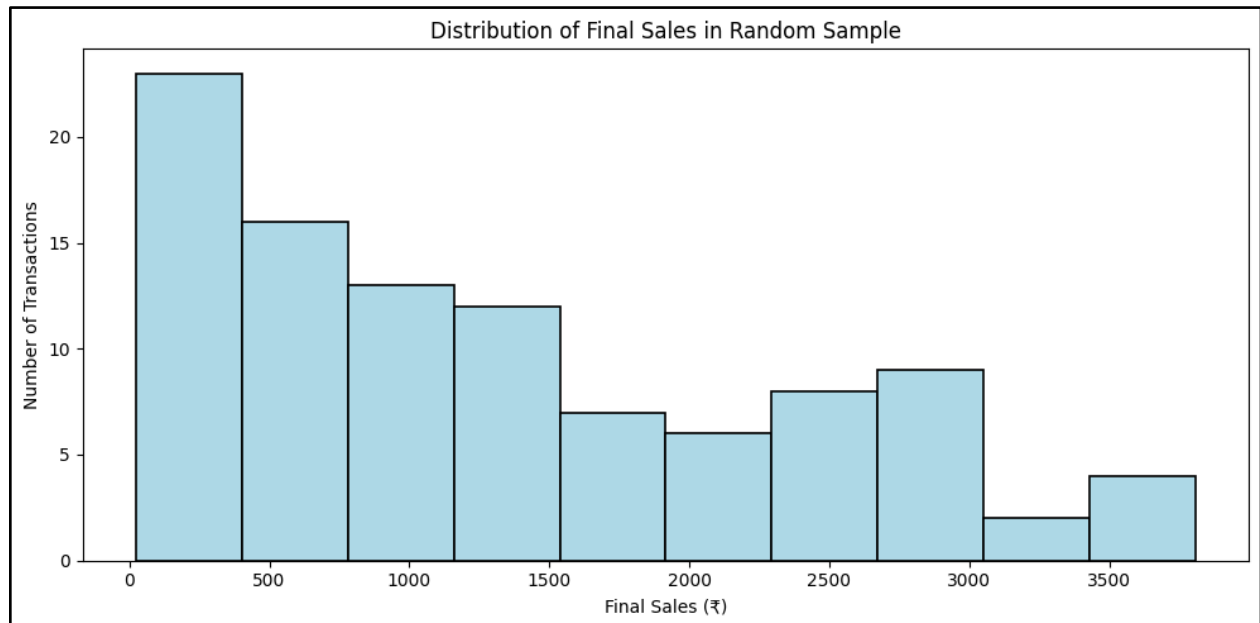
A sample size of 100 transactions was used for the analysis.

To understand sampling variability, multiple random samples were selected and their sample means were calculated. A sampling distribution was then created from the repeated sample means.

Sampling Details

- Population size: 1,000
- Sample size: 100
- Number of repeated samples: 100 or more
- Sampling method: Random sampling
- Measure analyzed: Final Sales

Figure 22: Sampling Distribution



Observation: The sampling distribution shows how sample means vary across repeated samples and provides an understanding of sampling variability.

17. Confidence Interval:

Confidence intervals were calculated to estimate the population mean Final Sales using sample data.

Three confidence levels were considered:

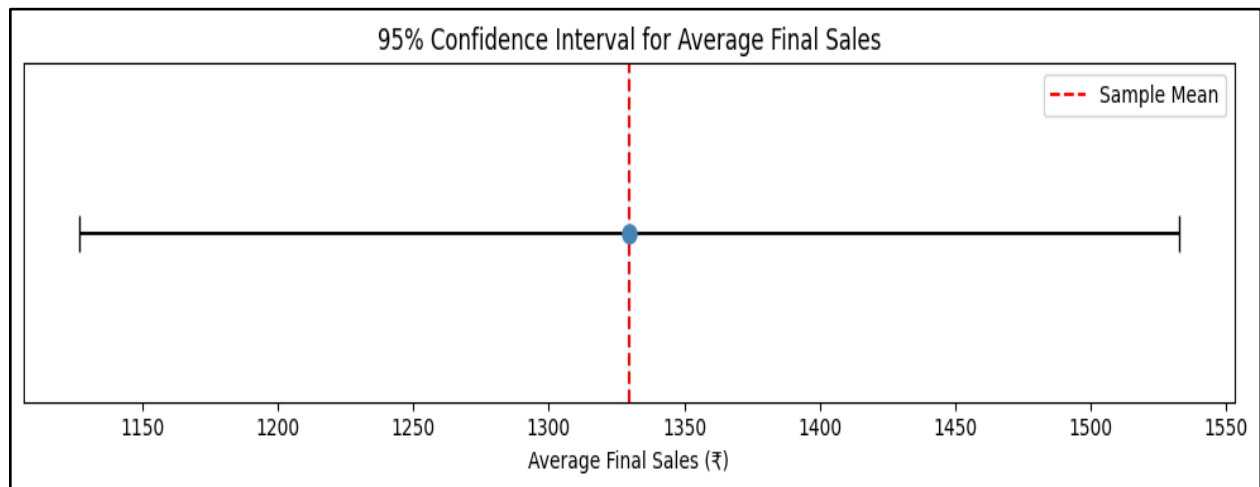
- 90%
- 95%
- 99%

Confidence Interval Table:

```
----- 95% Confidence Interval -----  
Sample Mean Sales: ₹ 1329.62  
Margin of Error: ₹ 203.17  
Lower Limit: ₹ 1126.45  
Upper Limit: ₹ 1532.79
```

A higher confidence level results in a wider interval because greater confidence requires a broader range of possible population values.

Figure 23: Confidence Interval Comparison



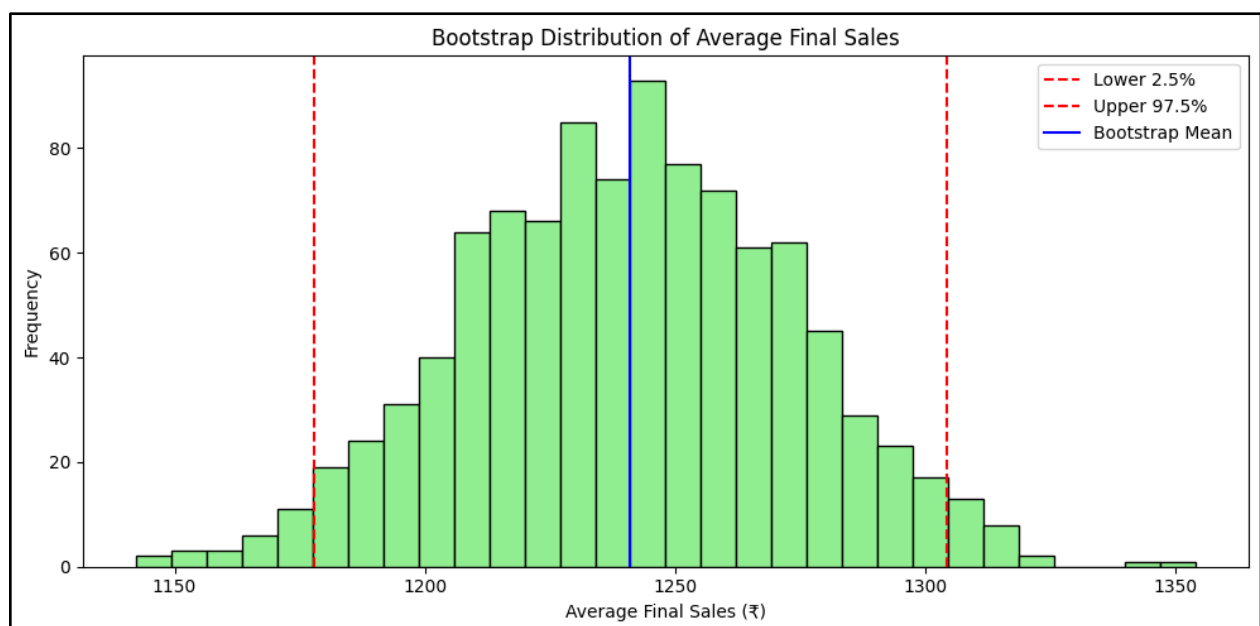
18. Bootstrapping:

Bootstrapping is a resampling technique used to estimate the uncertainty of a statistical measure.

In this project, 1,000 bootstrap samples were generated by sampling the available sales observations with replacement. The mean of each bootstrap sample was calculated.

A 95% bootstrap confidence interval was then obtained from the bootstrap distribution.

Figure 24: Bootstrap Distribution



Observation: The bootstrap distribution shows the variation of the estimated mean across repeated resamples and provides an estimate of the uncertainty associated with the sample mean.

19. Hypothesis Testing:

A one-sample t-test was used to determine whether the average Final Sales was significantly different from ₹1,200.

19.1 Hypotheses

Null Hypothesis (H_0): The average Final Sales is equal to ₹1,200.

Alternative Hypothesis (H_1): The average Final Sales is not equal to ₹1,200.

19.2 Significance Level: $\alpha = 0.05$

19.3 Test Used: One-sample t-test

19.4 Decision Rule: If p-value < 0.05, reject H_0 , If p-value $\geq$ 0.05, fail to reject H_0 .

19.5 Result

The calculated t-statistic and p-value were used to determine whether the observed average was statistically different from ₹1,200.

Figure 25: Hypothesis Testing Output

```
----- One-Sample T-Test -----  
Hypothesized Mean: ₹ 1200  
T-Statistic: 1.3202  
P-Value: 0.1871
```

19.6 Business Interpretation

The hypothesis test provides statistical evidence for determining whether the average Final Sales differs significantly from the assumed benchmark of ₹1,200.

20. A/B Testing:

A/B testing is a statistical method used to compare two alternatives and determine whether one performs better than the other.

For the stationery business, A/B testing can be used to compare two promotional strategies.

20.1 Alternatives

Group A: Existing promotional strategy

Group B: New promotional strategy

20.2 Metric

Conversion rate can be used to measure the percentage of customers who make a purchase after being exposed to the promotion.

Conversion Rate = Conversions / Total Users

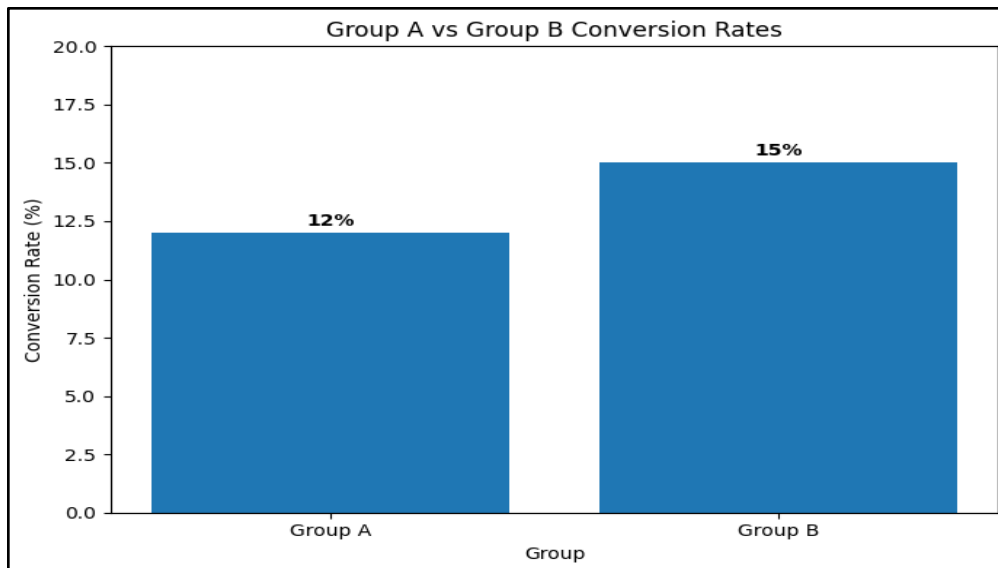
20.3 Hypotheses

H₀: There is no significant difference between the conversion rates of Group A and Group B.

H₁: There is a significant difference between the conversion rates of Group A and Group B.

A suitable statistical test can be applied to compare the two groups.

Figure 26: A/B Testing Comparison



Observation: The A/B test helps determine whether the new promotional strategy produces a statistically significant improvement in conversion rate.

21. Time Series Analysis:

Time Series Analysis was performed to examine sales behavior over time and forecast future sales.

The **Date** variable was used as the time variable and Final Sales was used as the sales measure.

21.1 Steps

1. Converted the Date column into datetime format.
2. Grouped transactions according to month.
3. Calculated monthly Final Sales.
4. Visualized monthly sales.
5. Identified sales trends.
6. Applied Exponential Smoothing.
7. Forecasted sales for the next three months.

21.2 Monthly Sales Findings

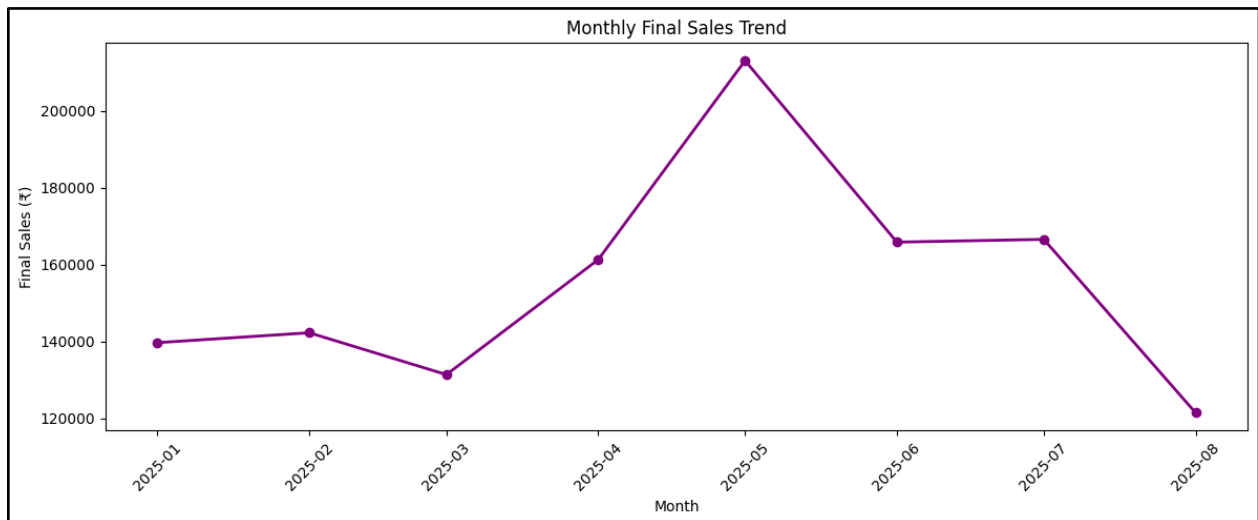
The highest monthly Final Sales was recorded in:

May 2025 – ₹2,13,082.22

The lowest monthly Final Sales was recorded in:

August 2025 – ₹1,21,439.66

Figure 27: Monthly Final Sales Trend



22. Sales Forecasting:

23. Forecast Accuracy:

Forecast accuracy measures were used to evaluate the performance of the forecasting model.

The following measures can be used:

Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

Mean Squared Error (MSE)

MSE measures the average squared difference between actual and predicted values.

Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and provides an indication of prediction error.

Forecast Accuracy Table

```
----- Forecast Growth -----  
September Forecast: ₹ 167666.32  
November Forecast: ₹ 173849.76  
Expected Growth: 3.69 %
```

Lower error values indicate better forecast performance.

24. Optimization Modeling:

Optimization modeling was used to determine the best production combination under limited machine and labour resources.

24.1 Decision Variables

Let:

A = Number of units of Product A

B = Number of units of Product B

24.2 Objective Function

The profit from Product A is ₹40 per unit and the profit from Product B is ₹30 per unit.

Therefore: **Maximize $Z = 40A + 30B$**

24.3 Constraints

Machine constraint: **$2A + B \leq 100$**

Labour constraint: **$A + 2B \leq 80$**

Non-negativity constraints: **$A \geq 0, B \geq 0$**

24.4 Optimal Solution

The optimal production plan is: **Product A = 40 units, Product B = 20 units**

24.5 Maximum Profit: Maximum Profit = ₹2,200

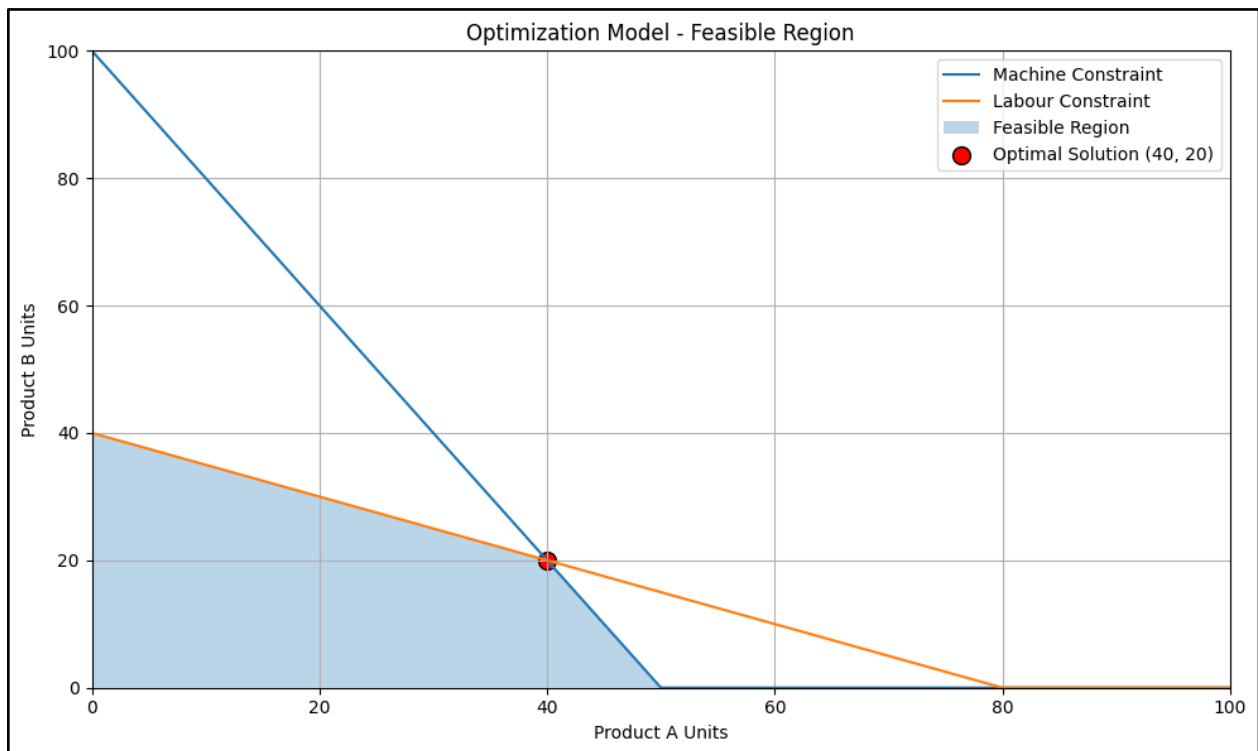
24.6 Resource Utilization

Machine: $2(40) + 20 = 100$ hours

Labour: $40 + 2(20) = 80$ hours

Therefore, both machine and labour resources are fully utilized.

Figure 29: Optimization Feasible Region



Observation: The optimal solution is 40 units of Product A and 20 units of Product B, producing a maximum profit of ₹2,200 while satisfying all resource constraints.

25. Simulation Modeling:

Simulation modeling was used to study possible business outcomes under different demand conditions.

Three demand scenarios can be considered:

1. Low Demand
2. Normal Demand
3. High Demand

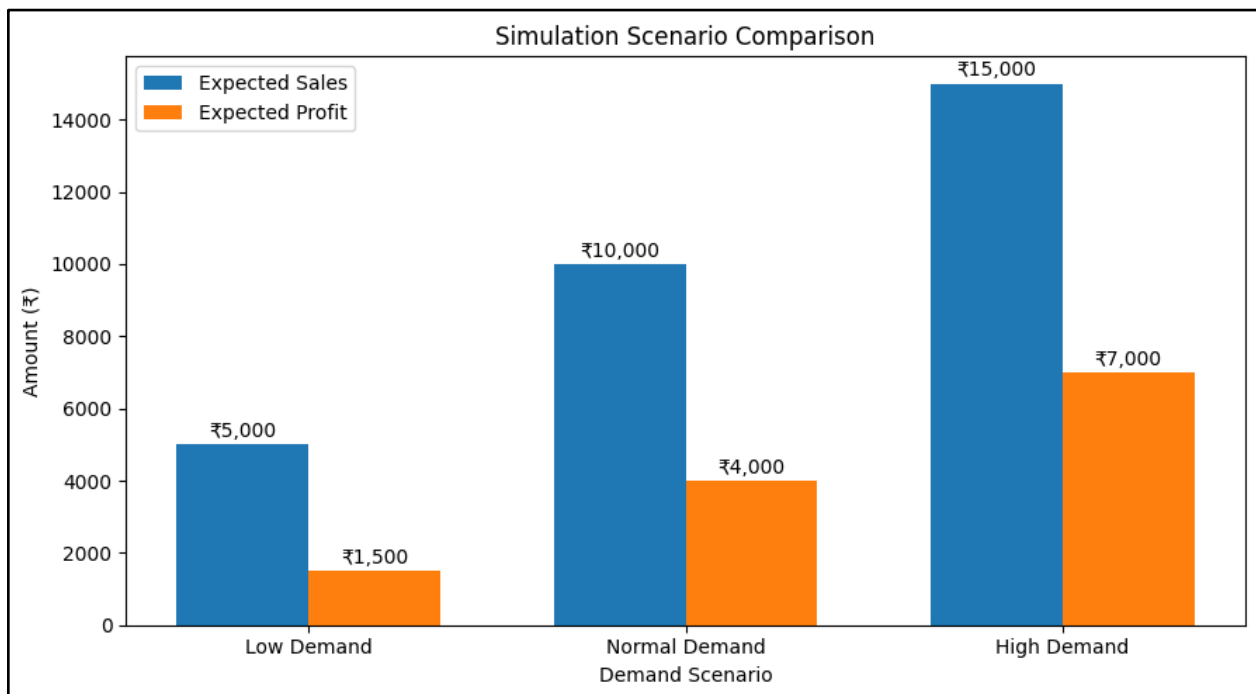
For each scenario, sales, profit, shortage and inventory levels can be simulated.

Scenario Comparison

Scenario	Demand Level	Expected Sales	Expected Profit
Low Demand	Low	₹5,000	₹1,500
Normal Demand	Medium	₹10,000	₹4,000
High Demand	High	₹15,000	₹7,000

The simulation allows the business to understand possible outcomes before making inventory or resource decisions.

Figure 30: Simulation Scenario Comparison



26. ANOVA:

Analysis of Variance (ANOVA) was used to determine whether there was a statistically significant difference in average Final Sales among different product categories.

26.1 Hypotheses

Null Hypothesis (H₀): The average Final Sales of all product categories are equal.

Alternative Hypothesis (H_1): At least one product category has a different average Final Sales.

26.2 Significance Level $\alpha = 0.05$

26.3 Results: F-statistic = 0.4855, p-value = 0.6155

26.4 Decision: Since the p-value is greater than 0.05, the null hypothesis is not rejected.

26.5 Conclusion

There is no statistically significant difference in average Final Sales among the product categories.

Figure 31: ANOVA Output

```
----- One-Way ANOVA Test -----  
F-Statistic: 0.4855  
P-Value: 0.6155
```

27. Experimental Design

Experimental design provides a structured method for studying how different factors affect a business outcome.

For Mayur Stationery, an experiment can be designed to study the effect of discount percentage and product category on sales.

27.1 Factor 1 – Discount Level

- Low Discount
- Medium Discount
- High Discount

27.2 Factor 2 – Product Category

Different product categories can be selected for the experiment.

27.3 Response Variable: Final Sales or Units Sold

27.4 Experimental Units: Individual customer transactions or selected sales observations.

27.5 Experimental Procedure

Different discount levels can be assigned to selected products or customer groups, and the resulting sales performance can be measured and compared.

The experiment can help determine which discount strategy produces better sales results.

28. Factorial Experiment

A factorial experiment was designed to study the effect of two factors simultaneously.

28.1 Factor A – Discount

- Low
- Medium
- High

28.2 Factor B – Product Category

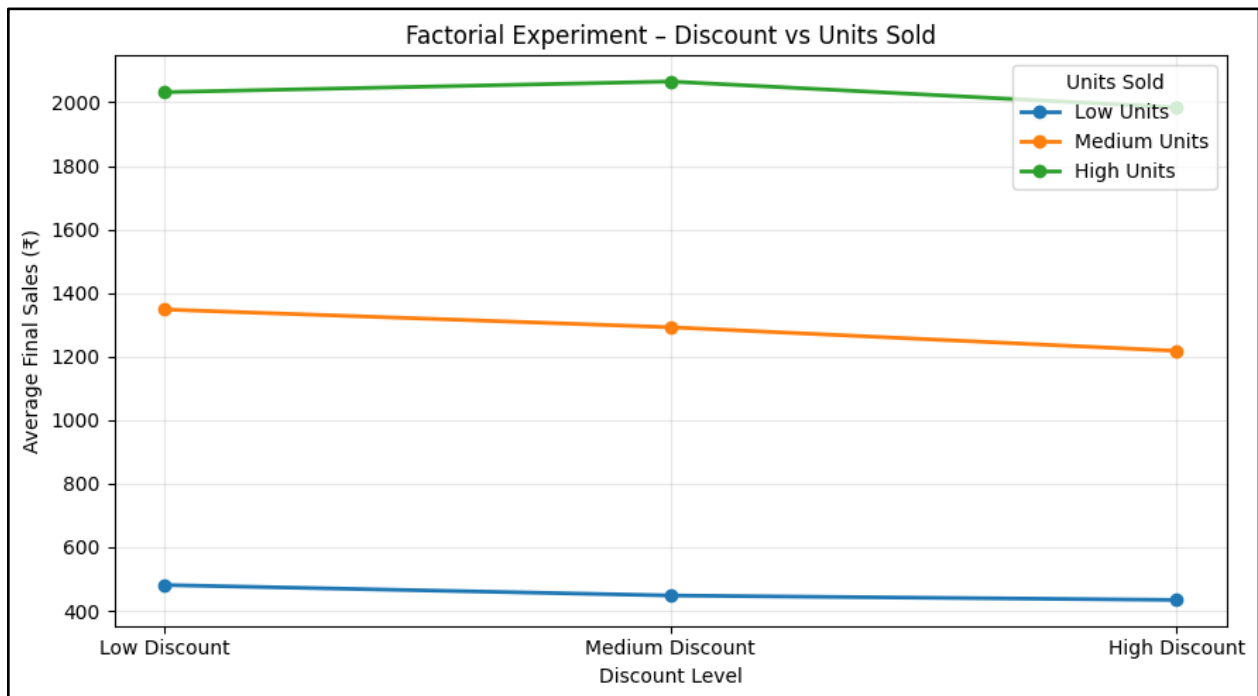
- Category 1
- Category 2
- Category 3

28.3 Response Variable: Final Sales

The experiment allows the following effects to be studied:

1. Main effect of Discount
2. Main effect of Product Category
3. Interaction effect between Discount and Product Category

Figure 32: Factorial Experiment Results



Observation: The factorial design helps determine whether discount levels, product categories and their interaction have an effect on sales performance.

29. Business Insights and Findings:

The major findings obtained from the analysis are:

1. The dataset contains 1,000 sales transactions and has no missing values or duplicate records.
2. Discounts are widely used, with 99.8% of transactions receiving a discount.
3. May 2025 recorded the highest monthly Final Sales of ₹2,13,082.22.
4. August 2025 recorded the lowest monthly Final Sales of ₹1,21,439.66.
5. The Exponential Smoothing model forecasts sales of approximately ₹1,67,666.32 for September 2025.
6. Forecasted sales increase to approximately ₹1,73,849.76 by November 2025.
7. The forecast indicates approximately 3.69% growth from September to November 2025.
8. The ANOVA test produced a p-value of 0.6155, indicating no statistically significant difference in average Final Sales among product categories.
9. The optimization model recommends producing 40 units of Product A and 20 units of Product B.
10. The optimal production plan produces a maximum profit of ₹2,200 while fully utilizing available machine and labour resources.

30. Business Recommendations:

Based on the analysis, the following recommendations are proposed:

1. Historical monthly sales should be used for inventory and stock planning.
2. Inventory levels should be increased during periods showing stronger sales performance.
3. Sales forecasting should be performed regularly to support future inventory and sales planning.
4. Since discounts are applied to almost all transactions, the business should evaluate whether the discounts are generating sufficient additional sales and profit.
5. Products and categories with stronger sales performance should receive appropriate inventory and promotional attention.
6. Store locations with higher sales should be analyzed to identify successful practices that can be applied to other locations.
7. Statistical analysis should be used before making major pricing and promotional decisions.
8. Optimization techniques should be used to allocate limited resources efficiently and maximize profitability.

31. Conclusion:

This Applied Business Analytics project analyzed the sales data of Mayur Stationery using statistical, analytical and predictive techniques. The dataset contained 1,000 transaction records covering products, customers, prices, discounts, categories, store locations and payment methods.

The project included data preprocessing, Exploratory Data Analysis, descriptive statistics, distribution analysis, correlation, regression, probability, sampling, confidence intervals, bootstrapping, hypothesis testing, A/B testing, time series forecasting, optimization, simulation and ANOVA.

The analysis identified important sales patterns and business findings. May 2025 recorded the highest monthly Final Sales of ₹2,13,082.22, while August 2025 recorded the lowest at ₹1,21,439.66. The Exponential Smoothing model predicted a gradual increase in sales from September to November 2025, with approximately 3.69% growth.

The ANOVA result showed that there was no statistically significant difference in average Final Sales among product categories. The optimization model showed that producing 40 units of Product A and 20 units of Product B could generate a maximum profit of ₹2,200 while fully utilizing the available machine and labour resources.

Overall, the project demonstrates how Business Analytics can transform raw sales data into useful information for business planning, forecasting, resource allocation and decision-making. The findings can help Mayur Stationery improve inventory planning, evaluate promotional strategies and make more informed business decisions.

32. Appendix:

Appendix A – Dataset & Preprocessing

- Dataset loading and basic information
- Missing values and duplicate checking
- Date conversion
- Data cleaning and Final Sales calculation

Appendix B – Data Analysis

- Descriptive statistics
- Correlation and regression
- Main EDA graphs and charts

Appendix C – Statistical & Probability Analysis

- Probability and sampling results
- Confidence interval
- Hypothesis testing and ANOVA

Appendix D – Time Series & Optimization

- Monthly sales and forecasting
- Forecast results
- Optimization code and solution
- Feasible-region graph

Appendix E – Dashboard

- Final ABA dashboard screenshot
- KPI results and major visualizations

Appendix F – Additional Analysis

- Decision tree
- A/B testing
- Simulation
- Experimental design
- Factorial experiment